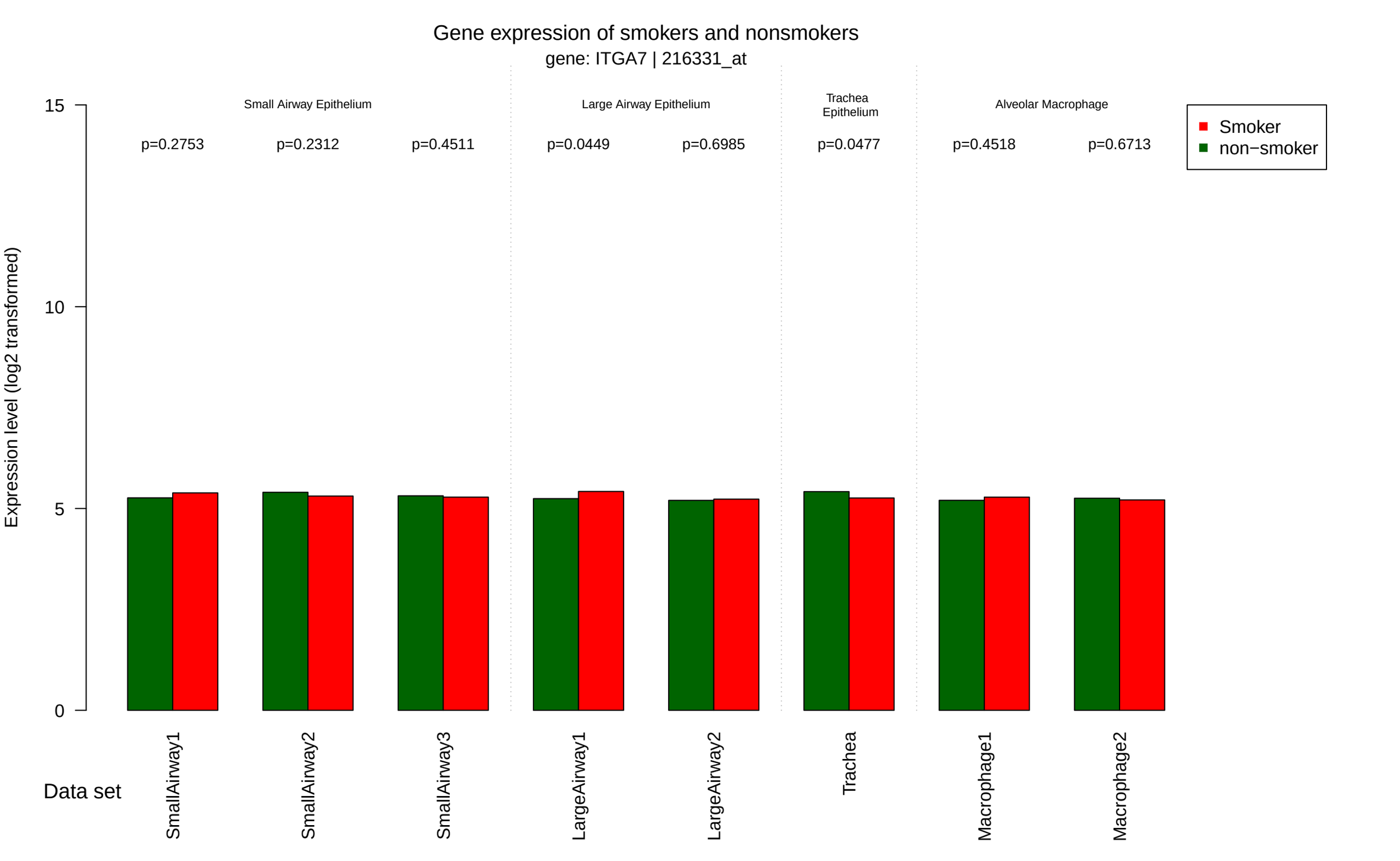
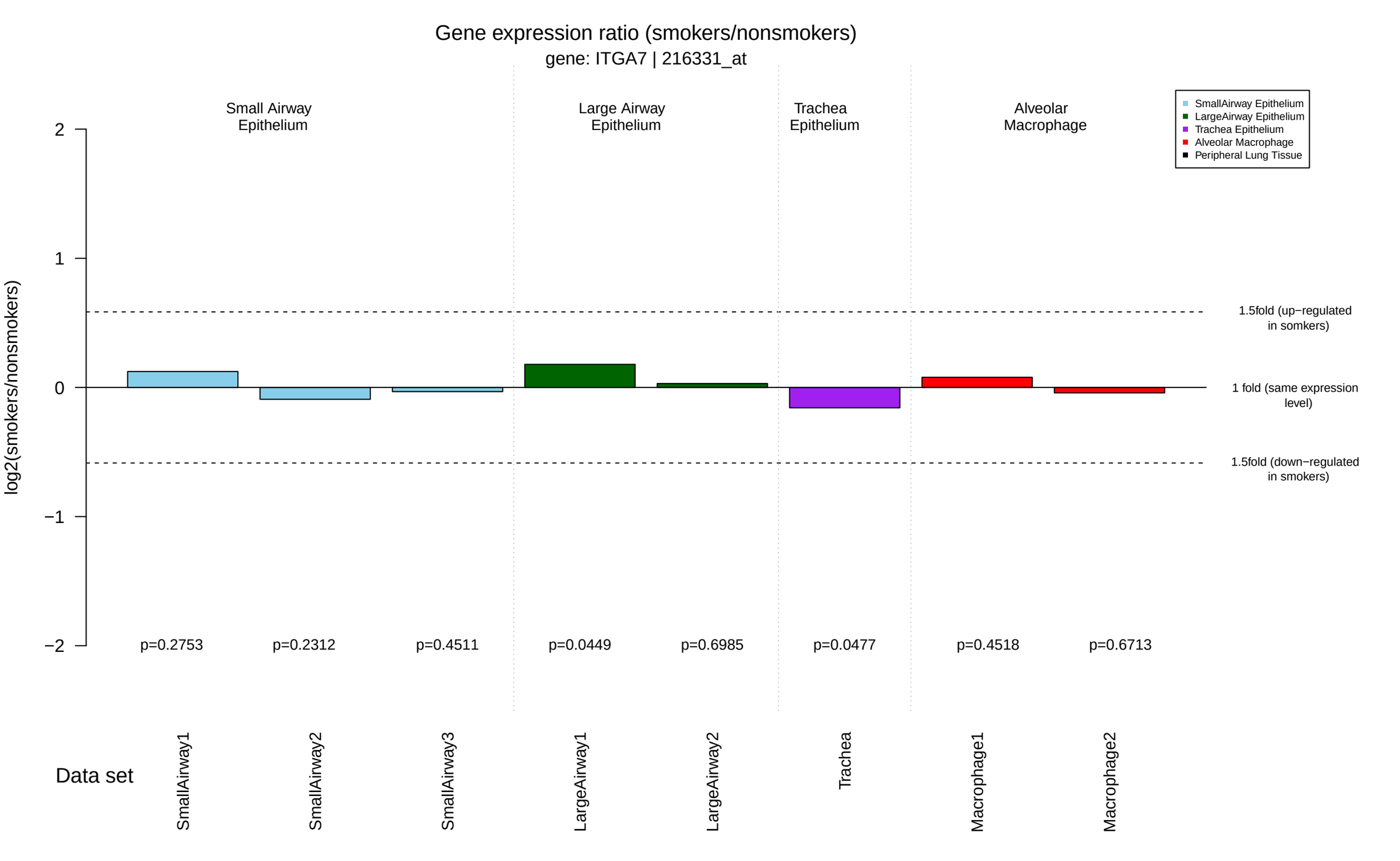
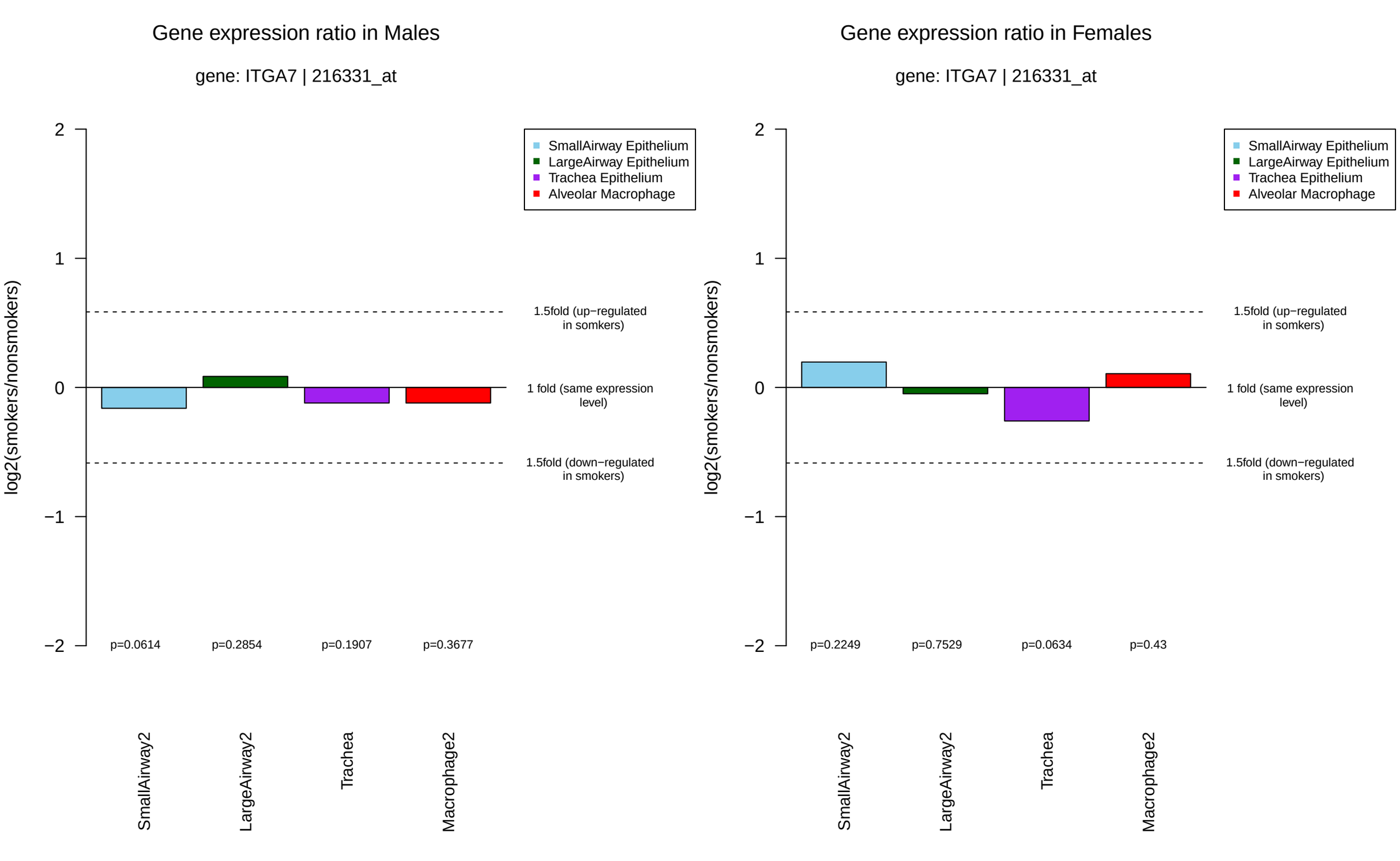
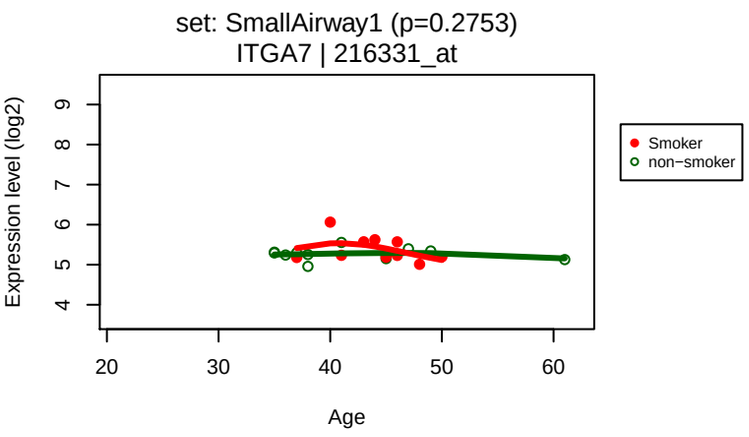
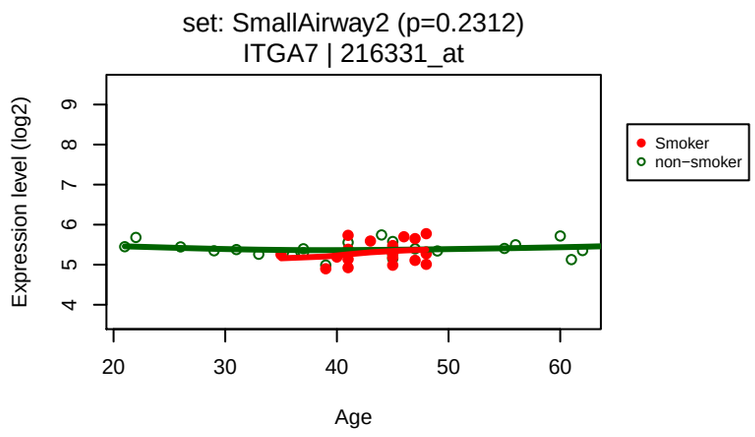
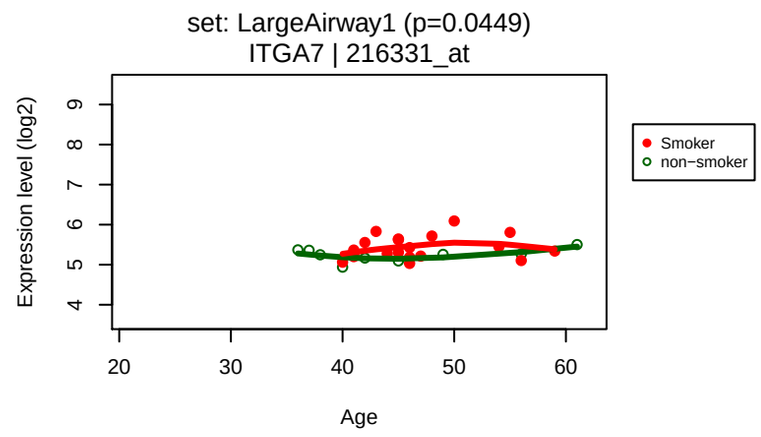

Gene expression of smokers and nonsmokers
gene: ITGA7 | 216331_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2753
p=0.2312
p=0.4511
p=0.0449
p=0.6985
p=0.0477
p=0.4518
p=0.6713
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

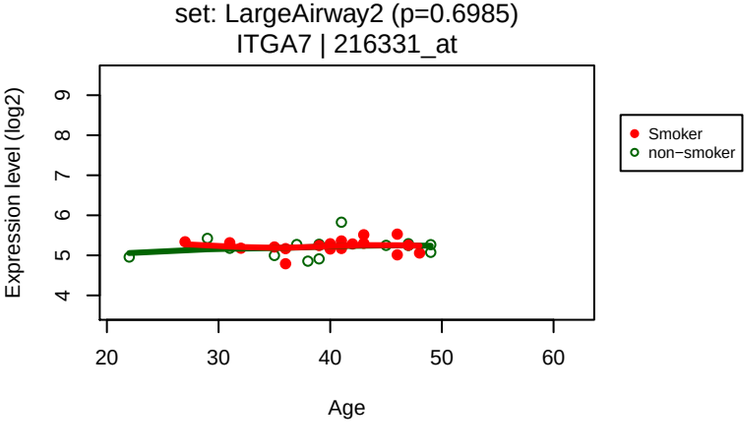

Gene expression ratio (smokers/nonsmokers)
gene: ITGA7 | 216331_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2753
p=0.2312
p=0.4511
p=0.0449
p=0.6985
p=0.0477
p=0.4518
p=0.6713
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

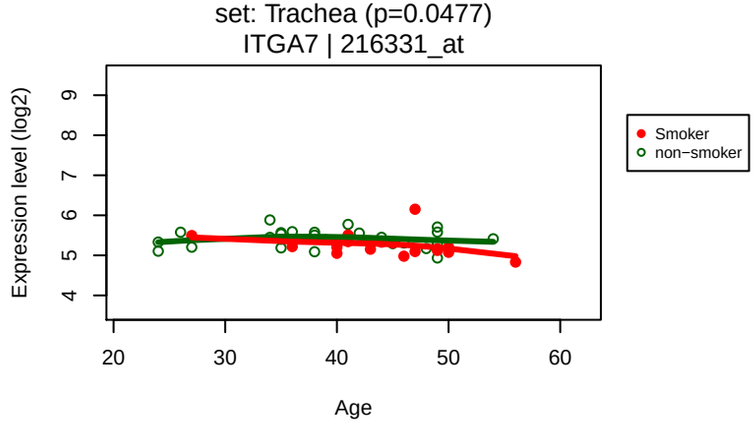

Gene expression ratio in Males
gene: ITGA7 | 216331_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0614
p=0.2854
p=0.1907
p=0.3677
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ITGA7 | 216331_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2249
p=0.7529
p=0.0634
p=0.43
SmallAirway2
LargeAirway2
Trachea
Macrophage2

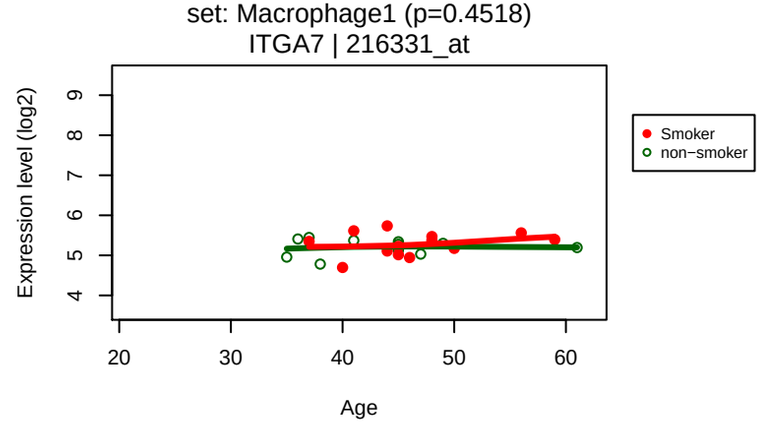

set: SmallAirway1 (p=0.2753)
ITGA7 | 216331_at
Expression level (log2)
Age
Smoker
non−smoker

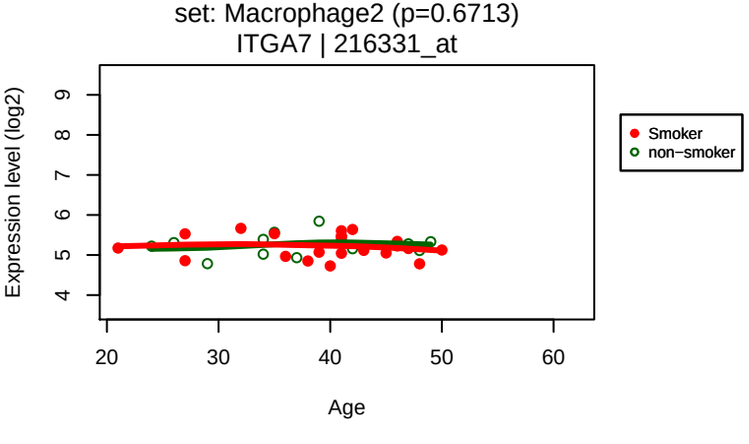

set: SmallAirway2 (p=0.2312)
ITGA7 | 216331_at
Expression level (log2)
Age
Smoker
non−smoker

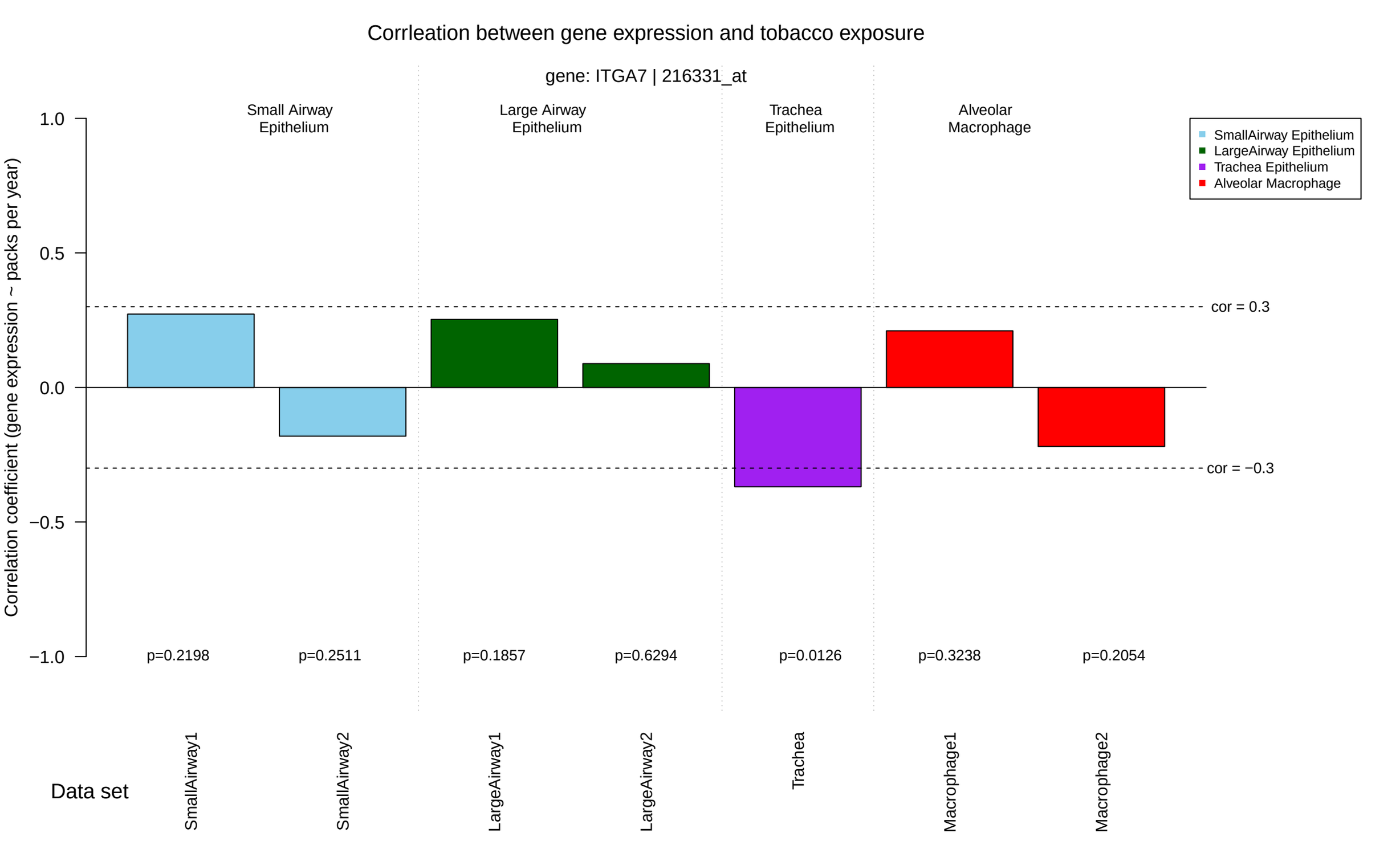

set: LargeAirway1 (p=0.0449)
ITGA7 | 216331_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6985)
ITGA7 | 216331_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0477)
ITGA7 | 216331_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4518)
ITGA7 | 216331_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6713)
ITGA7 | 216331_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ITGA7 | 216331_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2198
p=0.2511
p=0.1857
p=0.6294
p=0.0126
p=0.3238
p=0.2054
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2